S(20): DYADIC REDUCTION AND RELATIVE FORMAL IMMERSIONS

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Theorem. For elliptic curves over number fields of exact degree 20,
$S(20) = \{2,3,5,7,11,13,17,19,23,29,31,37,41,43,47,61\}$.
This is a computer-assisted proof. Every local exclusion, selected character family, independent rank and Pade check, and the complete prime and support coverage ledger has passed without an unresolved case.

1. EXISTENCE AND WEIGHTED MODULAR TRACES

For every prime through 7, a rational elliptic curve with the required point extends to a field of degree 20. Direct examples and the irreducibility of $t^{20}-2$ are checked in verify_small_occurrences.m. For $p=11,13,17,19$, $X_1(p)$ has a rational cusp and genus $1,2,5,7$. Both 19 and 20 exceed $2g-2$, so Riemann--Roch gives a function with its only pole of exact order 20 at that cusp. Hilbert irreducibility and the fine moduli interpretation give noncuspidal points of degree 20 and hence the desired elliptic curves.

Explicit witnesses for 23,29,31,37,41,43,47,61 are retained in occurrences.json and verify_occurrences.m. The latter proves the field polynomials irreducible, checks nonsingularity, and verifies that the specified nonzero point is killed by the prime. The first four fields have degree 10 and admit extensions of relative degree 2. The last four have degree 20. The 61 witness has $j=0$. The data were taken from local low-degree-place results, but only the validity of the individual witnesses is used.

Conversely, suppose E,P are defined over a field K of degree 20, with P of prime order $p>47$, $p\neq 61$. The associated modular point x has degree r dividing 20. Use the rational effective divisor
$D=(20/r)*\mathrm{Tr}_{Q}(x)$
on $X_1(p)$. It has degree 20, no cuspidal support, Galois-transitive geometric support, and constant geometric multiplicity. In particular, if a rational diamond h fixes D, all h-orbits on its support have the same length, dividing r and thus dividing 20. No assumption that x itself has degree 20 is made. All symmetric-power and principal-divisor tests below allow nonreduced divisors.

2. THE IMPROVED DYADIC BOUND

We use the previously established dyadic winding-quotient theorem in its integral form, with cotangent lattices satisfying
$2M \subset L \subset M$.
Equality of these lattices is not assumed. For $m=40=2*20$ and $M=229$, all 228 matrices
$(\varepsilon(na)-\varepsilon(nu/a))_{(n,a)}$,
$1\le n\le 40$, $1\le a\le 228$, $1\le u\le 228$,
have rank 40 over F_2. Division is modulo 229; ε is 1 on residues $115,\dots,228$ and zero on residues $1,\dots,114$. The retained matrices are independently reconstructed entry by entry and ranked in Sage. The resulting threshold is
$p>2*40*229=18320$.

The reduction kernel on rational dyadic torsion has exponent 2. Doubling the degree 20 trace therefore gives a degree 40 divisor whose image is zero. In symmetric cusp coordinates, if the divisor polynomial is $1+2B(t)$, its double has polynomial
$(1+2B(t))^2=1+4(B(t)+B(t)^2)$.

The doubled point is thus in the formal neighborhood modulo 4. Target functions have linear terms 2x_i and integral terms of degree at least 2. For a nonzero x in 4Z_2^40, let k>=2 be the smallest coordinate valuation. The corresponding linear term has valuation k+1<2k and cannot cancel. This proves that the zero fiber in this neighborhood consists only of the cuspidal point. The theorem excludes a trace whose specialization on X_0(p) is a multiple of one rational cusp, including ramified traces.

An ordinary elliptic curve over F_(2^f) has even group order. At f=20, its Hasse upper bound is 1050625; hence an odd prime factor is at most 525312. The complete supersingular orders for f=18,19,20 are
18: 261121,261633,262145,262657,263169;
19: 523265,524289,525313;
20: 1046529,1047553,1048577,1049601,1050625.
Every twist is retained and counted independently in Sage. The degree 20 orders have no prime factor above 100000. For f<=17, ordinary parity and the supersingular order lists give no prime above 100000; for f<=16 the Hasse bound alone suffices. Split and nonsplit multiplicative identity components have orders 2^f-1 and 2^f+1. The degree 20 toric orders likewise have only small prime factors. Additive reduction cannot support these large odd primes.

Consequently p>525313 would force every dyadic specialization onto a nonidentity multiplicative component and hence onto the same rational cusp of X_0(p), contrary to the dyadic theorem. Thus p<=525313. Above 100000 the complete parity/Hasse, supersingular, and toric screen leaves 444 primes. Ordinary orders in degrees 18,19,20 are all tested as even multiples of p; the degree 20 possibilities include 2p,4p,6p,8p,10p. The prime 524287 is included in the character screen: the special single-toric-place argument from degree 19 is not used in degree 20.

3. GLOBAL DIAMOND AND CHARACTER EXCLUSIONS

We use Derickx--Stoll, Prime order torsion on elliptic curves over number fields, Part I: Asymptotics, arXiv:2505.14109v1, Theorem 3.3, Proposition 4.1, Table 2, Remark 4.4 and Corollary 7.5. Table 2 is used only in its proved range p<100000. Rational 2-primary torsion is assumed cuspidal only in the proved range p<=157.

Write str(p) for the order of the group generated by strange characters. If p^2>(2^19/325)*20+1 and (p-1)/(2*str(p)) does not divide 20, Corollary 7.5 excludes p, except for its explicit even-degree CM possibilities. Formally apply it to the least divisor e of 20 at which p occurs newly; its size condition holds since e<=20, and a divisor that does not divide 20 cannot divide e. For all e|20 the only exceptional CM prime above 47 is 61, already in the lower bound.

Together with Proposition 4.1 and
gon_Q(X_1(p)) >= 325*(p^2-1)/65536,
these criteria leave 21 small residual primes:
53,59,67,71,73,79,83,89,97,101,103,107,109,113,127,131,137,139,
151,163,167.
The last three have global certificates.

At 151, diamond 2 has order 15. Its rational Jacobian image is finite, and the proved rational cuspidal 2-primary result permits Proposition 4.1 with ell=2, n=5: 100<gon(X_1(151)). Noncuspidal orbits have length 15 or 5. The remaining length 5 forces j=0. For the standard j=0 curve at q=5, Frobenius has characteristic polynomial X^2+5. The matrices

F^(6f)-1
are invertible modulo 151 for every 1<=f<=20, as checked exactly. This excludes the j=0 branch even over fields of degree dividing 20.

Here is the twist justification for this Frobenius test. For p>3 a nonidentity automorphism of an elliptic curve with j=0 or 1728 cannot fix a nonzero p-torsion point. Thus inertia of a twist must be trivial if such a point is defined over the local field. Twisted Frobenius is cF^f, where c belongs to the cyclic automorphism group of order w=6 or 4. Frobenius acts on this group by 1 or -1, and (cF^f)^w=F^(wf). Therefore invertibility of F^(wf)-1 excludes all twists. The analogous q=3, w=4 test at p=113 excludes the j=1728 branches used below.

At 163 put h=diamond 8, of order 27, and B=im(h-1). Its full rational homology consists of character orders 9,27,81, with coefficient degrees 6,18,54 and full sign-zero homology dimensions 24,24,26. Their total is
6*24+18*24+54*26=1980=2*990.
On every component A_3=T_3-diamond3-3 has odd integral determinant norm. The exact determinants and norms are independently checked in Sage. Thus A_3 on B is an odd-degree isogeny. Since (h-1)J(Q) is finite and killed by A_3, it has odd order. Proposition 4.1 with ell=2, a=8, n=6 applies since 120<gon(X_1(163)). The possible noncuspidal h-orbits have length 27 or 9, neither dividing 20.

At 167 use h=diamond 2 of order 83. B=im(h-1) has full sign-zero homology of dimension 26 over the degree 82 character field, hence rational homology dimension 2132=2*1066. An exact A_3 determinant has odd integral norm, independently recomputed in Sage. The same argument supplies the 2-primary hypothesis without a rational-torsion conjecture. Proposition 4.1 with n=5 gives invariance because 100<gon(X_1(167)); every noncuspidal orbit has length 83, which cannot divide 20.

For each of the 444 large primes set m=(p-1)/2. Choose a prime ell with v=v_ell(m)>w=v_ell(20), and screen every order n|m divisible by ell^(v-w). If all these orders are non-strange, then
v_ell(str(p))<=v-w-1,
so m/str(p) cannot divide 20. The size condition of Corollary 7.5 holds throughout this range. There are 3020 required character orders: 1069 inherited exact certificates and 1951 fresh calculations. Every family of divisors is checked explicitly; no table extrapolation is used.

The fresh calculations use modular symbols and the cuspidal winding element e'. For the k-dimensional cuspidal space, a scalar sequence u(T_2^i e') of length 2k has certified minimum recurrence length k. This proves that the winding element generates the entire space under T_2 and has nonzero projection on every relevant characteristic-zero factor. The Hecke action is the native Heilbronn action. All arithmetic is exact, the exported sequence checks native initial terms, and every 64-bit accumulation has an explicit bound. Independent Sage checks construct coprime Pade certificates C*S=A modulo X^(2k), with C(0)=1, deg(A)<k, deg(C)<=k and max(deg(A)+1,deg(C))=k. Coprimality excludes a shorter recurrence. Random choices find exact witnesses and are not a probabilistic assumption of the proof.

4. THE COMPLETE DYADIC CENSUS AND GLOBAL SECTION CERTIFICATES

For each of the remaining 18 levels, the finite-field census enumerates all j-invariants in degrees <=20, all twists, and every relevant cyclic p-torsion subgroup. When full E[p] is rational it includes all p+1 lines.

Proper j-fields are retained. Exact Frobenius orbits and complete diamond orbits distinguish ordinary, supersingular and nonrational cusp support. An additional independent census check uses Schoof’s class-number formula for ordinary elliptic curves, with class numbers enumerated unconditionally by reduced binary quadratic forms. Supersingular counts are independently obtained from the 24 Hurwitz units and the Frobenius coset i+j on F_p^2 minus zero. All counts agree in every degree 1,...,20.

The local census gives 370 support-degree multisets, including repetitions and a rational cusp complement to degree 20. It is reproduced directly in the final coverage script. Ordinary points have degree at least 7, so there are at most two ordinary components. At 73 three-component supports, including O7+O7+SS6, are explicitly included.

The degree 19 packet is inherited only after matching actual Tate normal-form diamond orbits. Its 11327-file manifest is unchanged. For 17 levels the canonical models are byte-identical to their inherited models. Level 139 has a fresh model of genus 737, precision 3600, and modular-unit pole degrees 253 and 265. All global q-series and unit identities pass. The integral cotangent lattice is proved saturated by the geometric zero bound for differentials: a minimal denominator would give a nonzero mod 2 differential vanishing beyond 2g-2. No optimal winding-quotient lattice equality is assumed. The verified canonical product dimension 3g-3 rules out a hyperelliptic special fiber.

All 22403 degree 20 ordinary sources at inherited levels and ordinary sources at 139 receive new jet and principal-divisor tests. Together with the matched inherited sources these cover 45391 ordinary orbits. Lower-degree ordinary sources and pairs are inherited through the actual orbit matches. Actual 3-isogenies provide the Hecke target records, whose weighted degrees sum to four times the source degree. Every jet equation and precision requirement is asserted. The finite-image annihilators are alpha*A_3, with alpha=1 at 59,71;
 alpha=(sigma-1)(sigma^6+1) at 97;
 alpha=(sigma-1)(sigma^5+1) at 101;
 alpha=sigma-1 otherwise.
The shorter valid polynomial at 97 kills its order 12 positive-rank character factor and permits the required degree 20 tests.

For a resulting rational divisor U-V over F_2 use L=K^2 at 53,67 and L=K at other levels. Choose a nonzero f in H^0(L(-U)). If U-V=div(u), then g=f/u is a nonzero section of L(-V), and h*g lies in f*H^0(L) for every h in H^0(L(-U)). The exact multiplication tensor and jet conditions therefore give necessary linear conditions on g. A full-rank matrix of these conditions proves nonprincipality. The sections and matrices are retained and independently ranked in Sage. There are 37905 new full-rank obstruction matrices, all independently verified.

The bicanonical choice at 67 is necessary: the ordinary degree 20 annihilator may have positive degree 160, larger than genus 155. Here dim H^0(K^2)=462 and dim H^0(K^4)=1078. Although precision 1300 is below the weight 8 modular-form Sturm bound, it exceeds the geometric bound for these particular sections: a product of four holomorphic differentials is a section of 4K of degree 8g-8=1232. Vanishing of its q-coefficients through precision 1300 gives differential vanishing order at least 1296>1232. All multiplication identities are therefore global.

Fresh pairs cover every relative diamond position of total degree 20; at 139 all pair totals <=20 are new. Mixed delta2 tests cover SS6+O14

at 73, NC11+O9 at 89, SS9+O11 at 109, SS7+O13 at 113 and NC7+O13 at 127. The added divisor is delta2 invariant, so only the ordinary portion enters the obstruction. The same observation applies to lower mixed tests, including SS6+SS6+O7 and NC9+O11 at 73. The new O7+O7+SS6 case at 73 has its own delta2 pair test in all 19 relative positions. These 21 additional matrices, including the two O11 sources at 109, have separate independent Sage rank checks.

For the actual elliptic 3-isogeny correspondence, the divisor convention is A_3=T_3-1-3*diamond3. On the q-form model its Fricke transport is T_3-diamond3-3 up to an invertible diamond factor. Thus the divisor tests and the cotangent tests below use the same torsion annihilator in their respective conventions. Such a factor does not change a kernel, cotangent rank, or isogeny-degree parity used in this proof.

5. ABSOLUTE AND RELATIVE FORMAL IMMERSIONS

Pure cusp disks use the integral DKSS endomorphism criterion, including its characteristic 2 form. Independence in the integral Hecke algebra is established by its matrices on modular symbols. Every raw operator row is reconstructed from the compression, so row compression is injective. Independence after a combined-map projection is used only in the direction implying independence of the actual operators.

The active finite-image polynomials are checked anew on J_0(p), using the correct transpose-dual Hecke action on period coordinates. Exact rational matrix ranks show their images lie in the rank-zero winding part. The proved strange-character table shows that no nontrivial positive-rank character descends to any active intermediate quotient. All eleven active polynomial checks have independent Sage rank checks. At 59 and 71 the full Jacobians are separately certified to have rank zero: their winding spans have the full dimensions 117 and 176, with both exact rational ranks independently checked in Sage. This justifies the rank-zero cases where arbitrary integral polynomials are used.

After moving a maximal cusp to position zero, complete dominating multiplicity profiles cover all distributions of degree 20. At the smaller indices, cap 6 patterns and every pattern with a second cusp of multiplicity >=7 are included. At larger indices the full relation spaces on cap 4 or cap 5 dominating profiles are computed. At 103 and 131 the only low-cost relations are the known full simple cusp fibers. Raw failures at the old 53 quotient and the old 101 index 5 map are not used; both are replaced by the full-curve maps described below.

The relative formal lemma is integral. Let S be smooth of relative dimension d over Z_2 and L a smooth closed formal family of dimension r. Suppose an actual integral map Phi:S->A to a smooth group scheme vanishes on L and has cotangent rank d-r. Choose normal parameters v for L. Suitable target functions lie in the ideal (v) and equal a matrix of power series times v, with invertible constant matrix. Their formal zero locus is exactly L. No integration denominator or quotient-cotangent saturation assertion is needed.

Our maps are actual integral endomorphisms of the Jacobian, transported by Fricke. Odd Hecke torsion factors kill their finite rational images and rational cusp base terms. Their cotangent matrices have the form
(D_h-1)(T_q-D_q-q),
or the explicitly stated sharper variants. Coefficient and trace-pairing matrices at a closed point differ by an invertible field-basis change.

If L contains a fixed nonzero cusp divisor R, a noncuspidal trace D cannot lie on L.

Canonical-coordinate ratios differ from differential coefficients by a common invertible local power series. Multiplication by that series gives an invertible triangular transformation on each finite jet space. Thus the stored normalized jets give the same vanishing conditions and cotangent ranks, including the first two jets at a repeated SS point. This uses no integration in characteristic 2. In the repeated SS73 case, the verified nonzero projective first derivative also certifies that the chosen elliptic deformation parameter is unramified on the modular curve, so it is a valid local parameter for this jet calculation.

At 53, SS13+C7 uses the free order 13 pullback and a fixed R7. The base rank is 12 and the relative family has dimension 1. Pure cusp disks are handled directly on the full curve by the order 13 diamond difference and the A_3,A_5 maps. All 63926 cap 3 profiles pass modulo the known fiber-jet relations. A degree 20 cusp divisor contains at most one full simple 13-fiber; extracting it leaves R of degree 20 or 7.

At 73 let H9=<2>. Its tame order 3 subgroup fixes the SS6 points with nontrivial tangent action, giving unique fixed lifts Z0. For SS6+C14 the family is Z0+R+pi^*Y, with zero or one cusp fiber and deg(R)=14 or 5. For NC9+C11 the NC divisor is itself a free H9 fiber; the base rank is 8 and extracting zero or one additional cusp fiber leaves R11 or R2.

The new double SS6+C8 disk is treated by relative immersion, not by an insufficient gonality inequality. At a repeated point the first two jets are obtained from a genuine elliptic deformation and verified against native division polynomials and all unit equations. The base rank is 12, for each of the three relative offsets and all four simultaneous translates. These 12 checks cover every unordered pair of the four closed SS6 points while the cusp normalization is fixed. On the symmetric square at a repeated tame fixed point the weights are zeta and zeta^2, so the fixed lift remains isolated. All relative cusp profiles have full normal rank 20, with fixed R8.

At 109, SS9+C11 uses the free order 9 subgroup, base rank 8, and a one-dimensional quotient pullback. Extracting zero or one cusp 9-fiber leaves R11 or R2. At 113, SS7+C13 uses the free order 7 subgroup, base rank 6, leaving R13 or R6. The old doubled gonality inequalities for these two disks are insufficient in degree 20; they are not used.

The exceptional O8+C12 disk at 113 uses the free order 8 action, base rank 7, and 392 profiles across all seven closed translates. A free action of the constant finite group near this orbit is an etale torsor even in characteristic 2. Thus the quotient pullback family is smooth without averaging invariant lattices. Extracting zero or one cusp 8-fiber leaves R12 or R4. The A_2,A_3 maps are justified by the proved rational cuspidal 2-primary result at 113.

The O9+NC7+C4 exception at 127 uses the product of the order 9 and order 7 diamond differences, with A_2,A_3. The same proved rational-torsion result applies at 127. The two quotient pullbacks give a smooth family of dimension 2 and base rank 14. Every one of 720720 cusp patterns has relative rank 18. The whole four-jet relation code has 252 columns and kernel dimension 60, generated by the two families of fibers. Each jet-level code is the 7-by-9 row/column code, with minimum nonzero Hamming weight 7; no four selected columns are dependent. The two

closed diamond-orbit lengths are 7 and 9, so the Chinese remainder theorem normalizes both supports simultaneously, and every cusp complement is included. The fixed R4 gives the contradiction.

At 103 and 131 use the odd-order quotient subgroups of orders 3 and 5. Their intrinsic invariant differential spaces have genera 120 and 131. All nonfiber cusp patterns pass absolute immersion. A full cusp fiber has degree 17 or 13; adding a fixed complement of degree 3 or 7 gives normal rank 19 in all 969 or 50388 patterns. The family is R+pi^*Y and has dimension 1. Pushforward of the weighted trace is allowed; a cusp in that pushforward forces a cusp in the original divisor.

All these cotangent maps, including every support and cusp entry, are independently rebuilt in Sage from the canonical model. Every retained quotient relation basis is checked to span the entire kernel modulo the known fiber relations. All its nonzero classes are enumerated and the minimum prefix cost is computed over every known-fiber mask. The cost exceeds the available degree, so only the stated pullback-family relations can occur. In total 105240 dominating profiles pass the independent relation-space checks.

6. SHARPER MAPS AT 97 AND 101

For p=97 and 101 the sole relevant nontrivial positive-rank character has order 12 and 10, respectively. For each n=2,3,5,7 the integral polynomial a_n(D) describing its Hecke eigenvalue is retained. Put

$$\beta_n = T_n - a_n(D).$$

The order 12 or 10 factor has dimension one over its degree 4 character field. An exact dual-period winding-image computation verifies that the image of each beta_n is in the rank-zero winding part. All eight number-field rank computations are independently repeated in Sage. Together with the proved strange-character table, this shows that

$$(h-1)\,\beta_n$$

has finite image on J(Q) when h kills the trivial character part. Composing with odd A_q therefore gives an actual rational-point annihilator, not a presumed optimal-quotient cotangent lattice.

At 97 take h of order 3, which fixes the SS16 geometric points. Their tame fixed lifts form Z0. For the sharper maps the SS base rank is 16 at both closed diamond positions. The C4 profiles pass modulo order 3 cusp fibers. Extracting zero or one such fiber leaves R4 or R1; the relative family Z0+R+pi^*Y has the required normal rank. The old direct principal-divisor rank 344/345 calculation is unused.

At 101 take h generating the full diamond group of order 50. The same sharper maps give 197 complete cusp profiles of full rank 20 modulo the full-group fiber-jet relations. No 50-fiber fits into degree 20, so every pure cusp disk is excluded. This replaces the unsuccessful index 5 cusp calculation. The actual Hecke matrices are reconstructed from full-rank q-expansion truncations, and every sharper cotangent entry is independently rebuilt.

7. STABILIZERS AND ALL REMAINING DISKS

Write Z=D-C, where C is the fixed rational cusp complement, and let

$$y=[hZ-Z].$$

This is a degree zero divisor class even when deg(Z)>0. If h has order m and N_h=1+h+...+h^(m-1), then

$$m*y=(h-1)[mZ-N_h\,Z].$$

The right-hand image is finite, so y is torsion. Special-fiber invariance gives reduction zero, hence 2y=0. If
$\mathrm{gon}(X_1(p))>2*(20+\deg(C))$,
comparison of the effective parts of 2(hD-D-hC+C) forces hD-D=hC-C. Their noncuspidal and cuspidal supports are disjoint, so hD=D.

At 73 a fresh export uses the actual JOne(73) integral homology of rank 374, with both Betti signs. The A_3,A_5 kernel intersection modulo 2 has dimension 6 and bounds the actual rational J[2]. Diamond 2 fixes this entire upper bound, as independently checked in Sage. Since $N_h*y=0$ and m=9 is odd, y=0. At 89 fresh exact calculations on both Betti signs give odd determinants on the order 11 anti-Jacobian, of dimension 260. The computed $T_3-1-3*D_3$ differs from T_3-D_3-3 by $2*(1-D_3)$. Both are integral endomorphisms preserving the actual anti-Jacobian, so their degree parities agree on its integral homology. Fricke transport preserves parity up to an invertible diamond. The proper torsion annihilator therefore has odd degree and kills the relevant rational 2-torsion, giving y=0. Both determinants are independently checked in Sage. These cases require only $\mathrm{gon}>20+\deg(C)$. All strict inequalities below are checked rationally.

At 73, nonrational total 18 with C2 and SS6+NC9 with C5 satisfy the undoubled inequalities 22 or 25<gon. H9-orbits have lengths 9 or 3, neither dividing 20. The double SS6+C8 disk uses the relative test of Section 5. At 89, NC11+C9 has 29<gon and H11 invariance, impossible for 20. At 109, SS12+C8 and total 18+C2 satisfy doubled bounds 56 and 44; H18-orbits have lengths 18,9,6, none dividing 20. SS9+C11 is relative.

At 113, total 14+C6 gives the doubled bound 52 and H14 invariance; lengths 14 or 7 do not divide 20. Pairs of exceptional O8 components have C4 and bound 48; H8-invariance could have orbit length 4 only at j=1728, already excluded by the Frobenius test. A single exceptional O8 is handled by relative immersion. For SS7+O8+C5 the doubled bound is 50, so diamond 98, an involution, fixes D. Its generic fixed points have j=1728 and are excluded. Otherwise D is a sum of free involution pairs, with the common multiplicity allowed by a weighted trace. Each pair specializes with even multiplicity at every fixed special-fiber point. But the SS7 component has multiplicity one at all its geometric points, which are fixed by the involution. This is impossible. No odd-degree or wild quotient-lattice argument is used.

At 127, NC7+C13 and total 14+C6 have doubled bounds 66 and 52, giving H7 invariance. Exceptional O9+C11 and pairs of exceptional O9 with C2 have bounds 62 and 44, giving H9 invariance. The possible orbit lengths 7, or 9 and 3, do not divide 20. The mixed exceptional O9+NC7 disk is relative. At 137, SS17+C3 and the inherited exceptional ordinary O17+C3 both have bound 46<gon. Their stabilizers have orders 34 and 17, with orbit lengths 34 or 17, incompatible with 20. All stabilizers are verified on actual Tate normal forms, without assuming their action from a residue-degree label alone.

8. COMPLETENESS AND REPLAY

Every one of the 370 local support-degree multisets has an explicit principal-divisor, absolute-immersion, relative-immersion, or stabilizer argument. Fresh actual-orbit matches connect inherited certificates to the new representatives. The global ledger checks every prime up to 525313 and every selected character-order family. The ledger has passed for all 43465 primes in the finite range and all

370 local support types. This proves the theorem.

REPLAY.txt records the verification commands. The certificate index links the exact models, witnesses, source jets, matrices, independent checks and coverage ledgers. The final manifest includes every frozen parent proof input and the new proof inputs. Unused exploratory failures are identified above and are not premises of the proof.

References: the previously proved integral dyadic theorem and inherited audit; Derickx--Kamienny--Stein--Stoll, Torsion points on elliptic curves over number fields of small degree, Algebra & Number Theory 17 (2023), Corollary 5.2 and Proposition 5.3; Derickx--Stoll, arXiv:2505.14109v1, Theorem 3.3, Proposition 4.1, Remark 4.4, Table 2 and Corollary 7.5; Schoof, Families of curves and weight distributions of codes, Proposition 5.2, for the independent ordinary point-count check.

9. EXACT GRAPH CERTIFICATES FOR THE FRESH CHARACTER CALCULATIONS

The faster character implementation uses a larger sparse representation, with a verified quotient argument; it does not infer winding generation from an arbitrary scalar sequence on that larger space.

Fix a nontrivial even character chi of prime conductor p, and an auxiliary prime q congruent to 1 modulo p-1. Let S be the F_q-vector space on Manin symbols after imposing the two-term S and complex-conjugation relations with sign +1. The native two-term quotient algorithm fixes its basis. Let A have as rows every three-term Manin relation in that basis, with proportional duplicates removed. Thus the actual relative modular-symbol space is exactly M=S/row(A). The raw operator R is the four-term Cremona Heilbronn action of determinant 2, followed only by the two-term reduction. It induces the usual T_2 on M; equivalently, it preserves the remaining Manin relation subspace. This is the standard Manin-symbol presentation and Hecke action, with the same character convention as the native code. The fast implementation caches the native character value list and uses the same projective-line reduction, rather than changing any relation.

Every column of the retained A has at most two nonzero entries; this is asserted on each actual matrix. Regard its rows as vertices. A two-entry column is a weighted edge; a one-entry column joins its vertex to a dummy anchor, and a zero column is free. A spanning forest provides one leaf pivot for every tree edge. In each component without an anchor, choose nonzero weights lambda satisfying

lambda_v*a_(v,e)+lambda_w*a_(w,e)=0

on all tree edges, with lambda=1 at the root. A nontree edge has reduced coefficient gamma_e=sum_v lambda_v*a_(v,e). If some gamma_e is nonzero, that chord provides the remaining root pivot. If all gamma_e vanish, lambda is an explicit left-null vector. These left-null vectors have disjoint nonempty support in the balanced components and are independent. Consequently the lower rank from the tree and chord pivots equals the upper rank from these left-null vectors. Both certificates are checked independently, for every actual character matrix, in verify_graph.py. This proves rank(A) exactly, not a probabilistic lower bound.

A column functional u is constructed with A*u=0. All of these equations are checked independently in Python and again in Magma. Write e_0 for the identity Manin symbol representing {0,infinity}, and set

e’=e_0*(R-(2*chi(2)+1))*(R-(chi(2)+2)).

For a nontrivial character of prime conductor, the boundary has dimension 2, and T_2 has precisely these two Eisenstein eigenvalues on it. Hence

the image of e’ lies in the cuspidal subspace of M, of dimension
 k=dim(S)-rank(A)-2.
The scalar sequence e’*R^i*u therefore factors through the actual k-dimensional cuspidal modular-symbol space, even though R acts on a larger ambient space. Its certified minimum recurrence length k proves full winding generation exactly as in Section 3. The scalar functional need not be chosen canonically: all its relation equations are exact certificates, and randomness only searches for a convenient witness.

The field q is prime, splits the character values, and is coprime to 6p(p-1). The character summand of relative homology and its boundary therefore have the usual good integral specialization. A full-rank winding witness modulo q lifts to characteristic zero. No conjecture on analytic ranks or rational torsion is used in this step.

Each fresh graph job retains the complete A and R matrices, the spanning forest and chord rank certificate, the functional, and the prepared winding vector. The complete input files are compressed losslessly; their compressed and decompressed hashes are checked separately. Magma independently reconstructs e’, checks all functional relations, and verifies 64 scalar terms before the C++ sequence is accepted. The C++ accumulation bound uses the larger actual ambient dimension. Jobs for which this bound would fail are routed to the original smaller native quotient representation; no overflowing calculation is used.

For the actual required character p=525253, order 1014, the graph has 131314 generators, 87541 independent three-term relations, and quotient dimension 43773, agreeing with the native full quotient. All 131314 raw Hecke rows agree with the native construction. Using the same descended functional, all 87542 sequence terms agree with the original quotient sequence, not merely the initial 64 terms. Every byte of the fast exported graph also agrees with the slower native graph construction. These comparisons are retained as implementation audits; the quotient, rank, functional and Eisenstein arguments above justify the method for each character on its own exact data.